

Supplementary Material

Table S1. Parametrization of algorithms.

Algorithm	Parametrization
PCA	n_components=2
MDS	n_components=2
ICA	n_components=2, function=logcosh, tolerance=1e-3
KPCA	n_components=2, kernel=rbf, gamma=0.1
SOM	size=10, sigma=1.0, lr=0.5, epochs=1000
AE	n_components=2, hidden_layers = [70,40,20,10], epochs=100, lr=1e-3, batch=64, activation=tanh
LLE	n_components=2, n_neighbors=70
MLLE	n_components=2, n_neighbors=50
Isomap	n_components=2, n_neighbors=100, eigensolver=Arnoldi decomposition, path=Dijkstra
Spectral embedding	n_components=2, affinity: nearest_neighbors
t-SNE	n_components=2, perplexity=30
Diffusion Map	n_components=2, alpha=0.5, k=50
PHATE	n_components=2
UMAP	n_components=2, n_neighbors=10, min_dist=0.05, metric=Chebyshev
Trimap	n_components=2

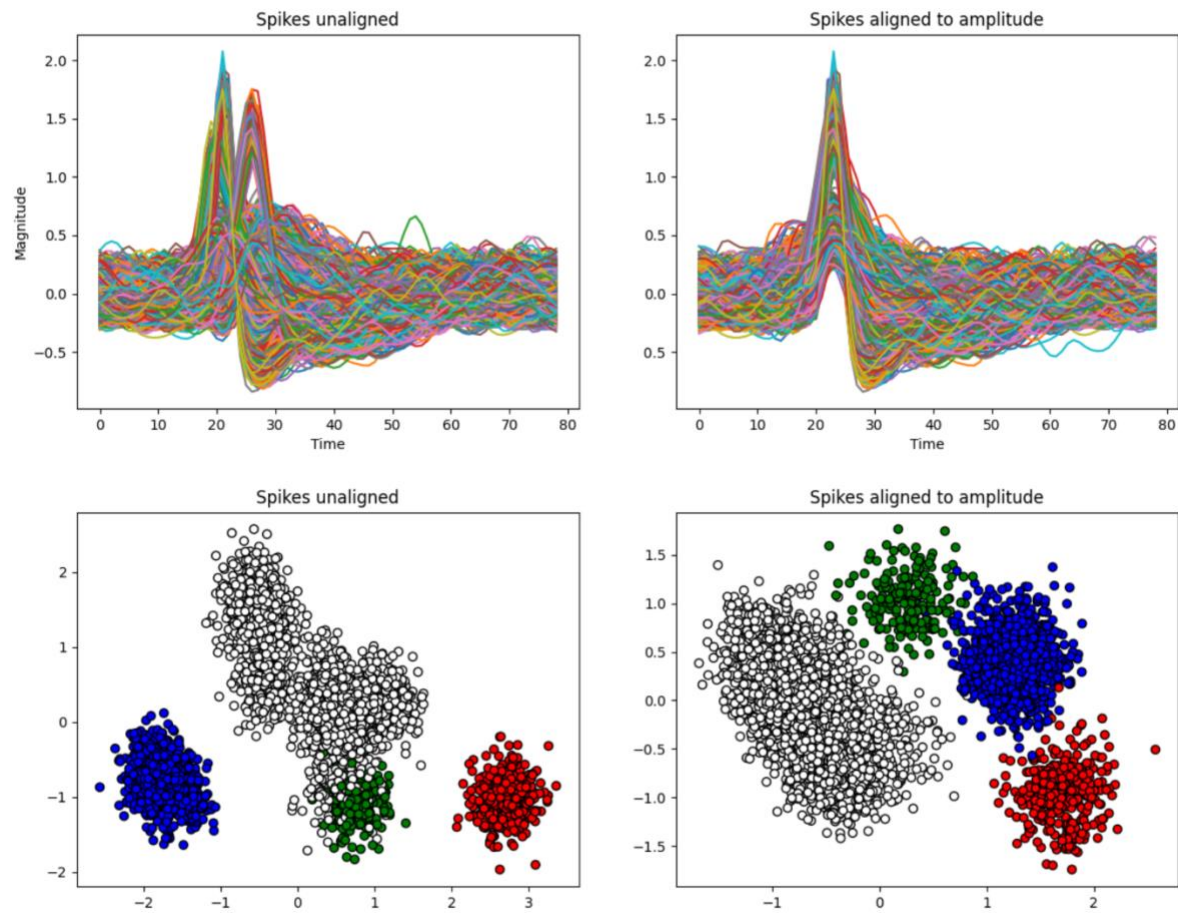


Fig S1. Impact of alignment on Sim53. The top row presents the impact of alignment on the spikes themselves, while the bottom row presents the impact of alignment on the PCA features. The white cluster is kept together and the green cluster is now separated from the white cluster.

Table S2. Information about the 95 synthetic datasets (19).

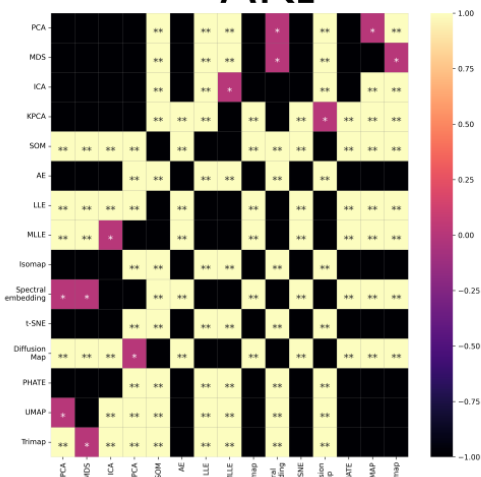
Simulation number	Number of ground truth clusters /units	Number of spikes
1	16	12012
2	19	12784
3	12	9277
4	4	5127
5	15	11758
6	11	10186
7	17	13999
8	2	3494
9	19	15653

10	20	15149
11	20	14982
12	20	13488
13	10	7792
14	3	4507
15	9	9683
16	8	7556
17	9	9098
18	7	7004
19	18	13807
20	14	11186
21	4	4293
22	6	7101
23	17	13374
24	6	6302
26	13	12313
28	14	12081
29	3	4177
30	5	5210
31	11	8224
32	10	9078
33	4	4169
34	8	6851
35	12	9481
36	18	12377
37	10	9954
38	13	11148
39	2	4719
40	7	7360
41	15	13347
42	9	6746
43	16	10871
45	8	6666
46	2	4202
47	15	10977
48	7	6823
49	11	9496
50	14	11211

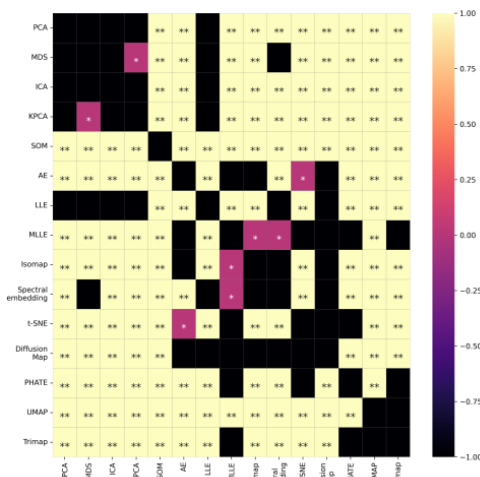
51	18	11746
52	12	9255
53	3	4490
54	5	6923
55	19	14117
56	17	12784
57	6	6301
58	7	7860
59	2	4176
60	12	10158
61	17	13280
62	6	6649
63	13	7634
64	4	4890
65	17	13523
66	14	12572
67	13	11377
68	9	7170
69	15	12051
70	11	9395
71	19	13459
72	16	11183
73	9	8436
74	20	13065
75	19	13908
76	3	4791
77	5	7084
78	10	8518
79	20	14536
80	10	7757
81	8	7937
82	14	10430
83	4	5206
84	6	5809
85	18	14512
86	18	13847
87	5	6671
88	8	6414

89	3	4573
90	12	10565
91	16	10821
92	11	8184
93	7	6113
94	2	4990
95	15	12462

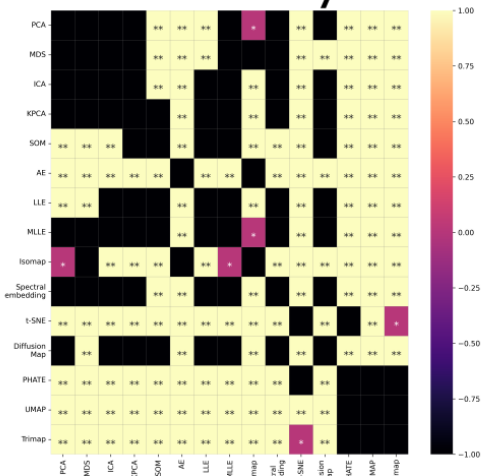
ARI



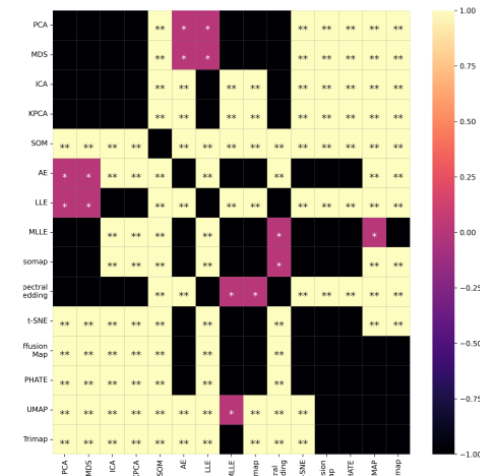
AMI



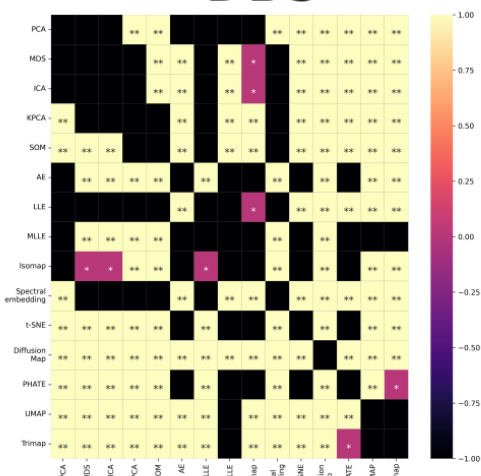
Purity



SS



DBS



CHS

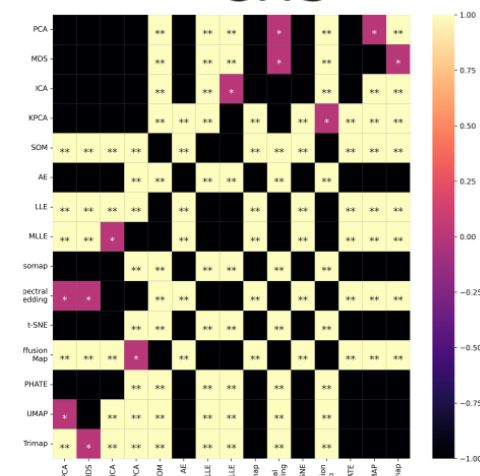


Fig S2 - P value of Mann–Whitney U tests (with a Bonferroni correction) for each of the metric on all 95 simulations (** represents $p < 0.01$, * represents $0.01 < p < 0.05$, while no text represents $0.05 < p$).